

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2

19 June 2026

Subject	Mathematics
Grade	12
Marks	150
Time	3 Hours

Examiner

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Moderator

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Mrs S Rowell

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 13 questions, 16 pages and a formula sheet. You may detach the formula sheet for your use.
Please ensure that your question paper is complete.
2. Answer ALL the questions in the answer books provided. Additional space is provided at the end of the answer book.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. Use the mark allocation to guide the length of your solution.
9. Write neatly and legibly.

QUESTION 1

The points of the top Rugby 7s teams on the worldwide rankings were as follows:

Ranking of the top 10 Rugby 7s	Points
South Africa	38
Argentina	34
Australia	30
Fiji	26
Spain	26
New Zealand	26
France	14
Kenya	14
Uruguay	6
Germany	5

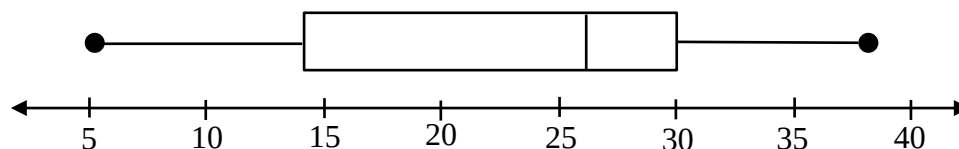


1.1 Calculate the average points (mean) of the top 10 teams. (1)

1.2 Determine the standard deviation of the points. (1)

1.3 Determine the number of teams whose points lie within one standard deviation of the mean. (3)

1.4 The data is represented in the box and whisker diagram below:



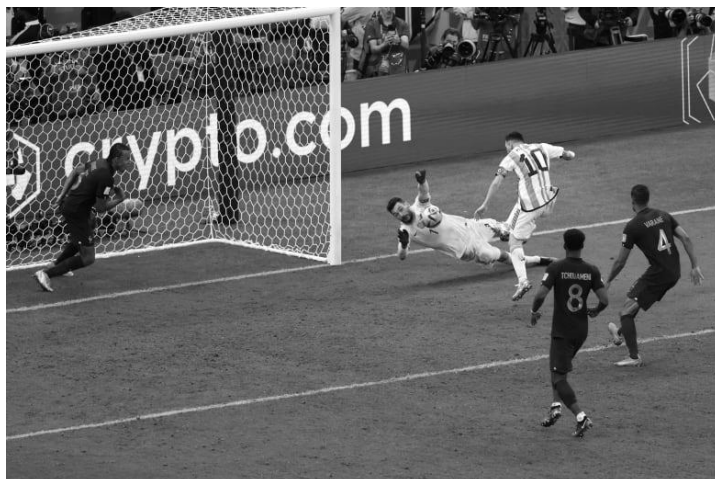
1.4.1 Comment on the distribution of the data. Provide a reason for your answer. (2)

1.4.2 Which one of the mean or median is the best measure of central tendency for this data? (1)

[8]

QUESTION 2

A soccer coach is considering a strategy of having defensive players taking more shots on goal. The coach evaluated a sample of the team's recent matches to see how players in defensive roles could contribute offensively. In each match the coach tracked the number of shots on goal by the defenders.



The coach recorded the differences between the number of goals that had been scored by the team and the opposing team in the table below:

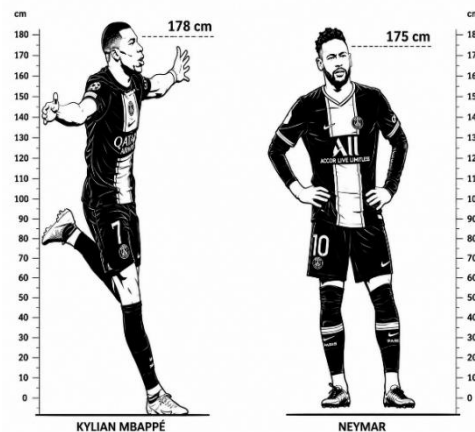
Shots on goal by defenders (x)	2	5	7	7	10
Goal difference (y)	-1	0	-1	0	3

- 2.1 Calculate the correlation coefficient of the data and comment on the relationship between the number of shots on goal by defenders and the goal difference. (2)
- 2.2 Determine the equation of the least squares regression line. (3)
- 2.3 Use the answer to QUESTION 2.2 to determine the estimated average change in the goal difference for each additional shot on goal by defenders. (1)

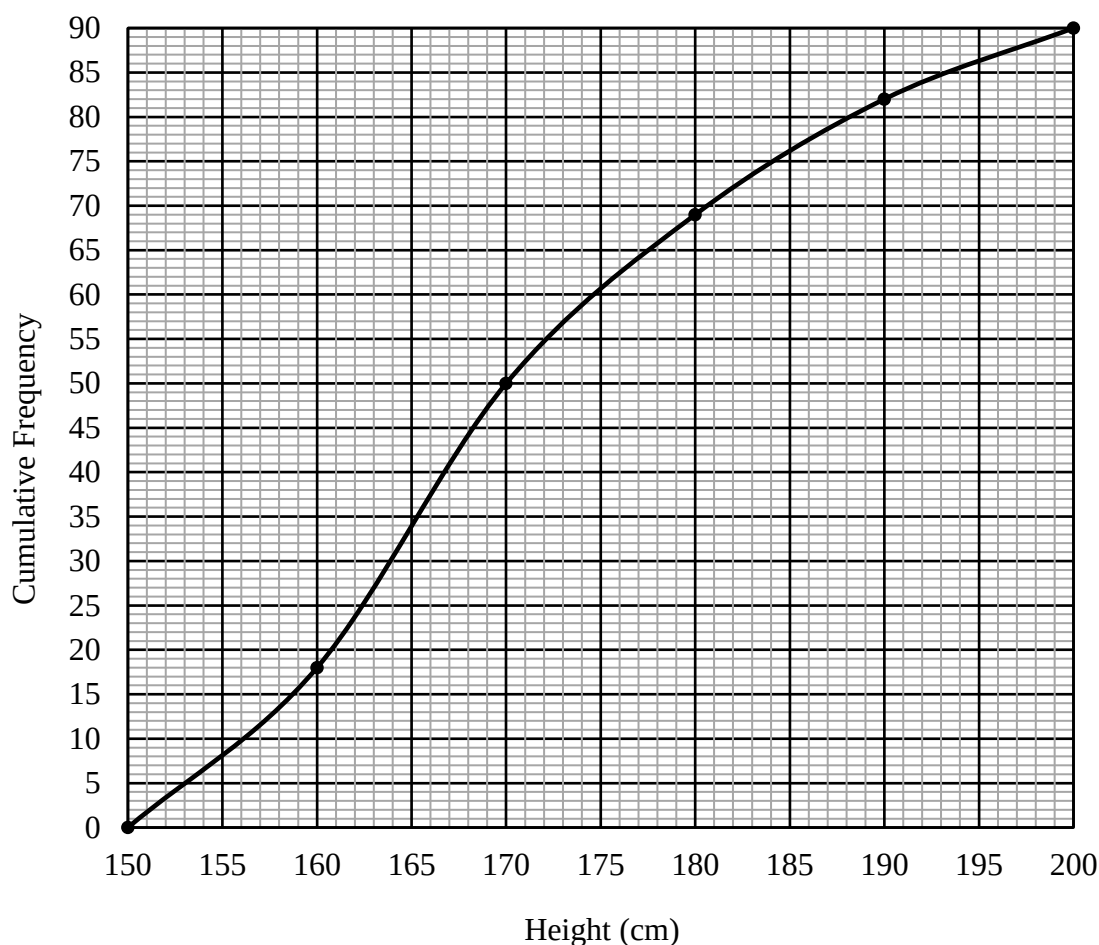
[6]

QUESTION 3

Height influences a professional soccer player's style. Shorter players benefit from a lower centre of gravity, making them ideal for dribbling in tight spaces. Taller players, with longer strides and reach, excel in aerial duels, and goalkeeping. Goalkeepers are usually the tallest in a team, averaging between 188 cm and 196 cm in height.



The results of a study on the heights of a group of soccer players are shown in the cumulative frequency graph (ogive) below.

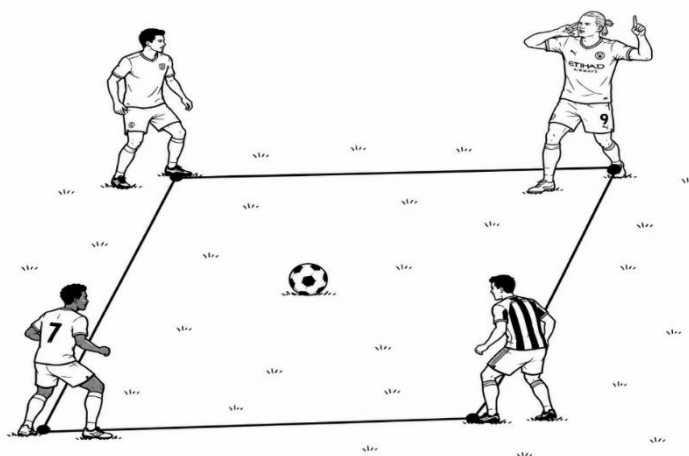


- 3.1 How many soccer players participated in the study? (1)
- 3.2 Use the graph to estimate the median. (1)
- 3.3 Use the graph to estimate the interquartile range. (3)
- 3.4 The average height of a male professional soccer player is about 1,8 m. What percentage, to the nearest integer, of the soccer players in the study are taller than 1,8 m? (2)

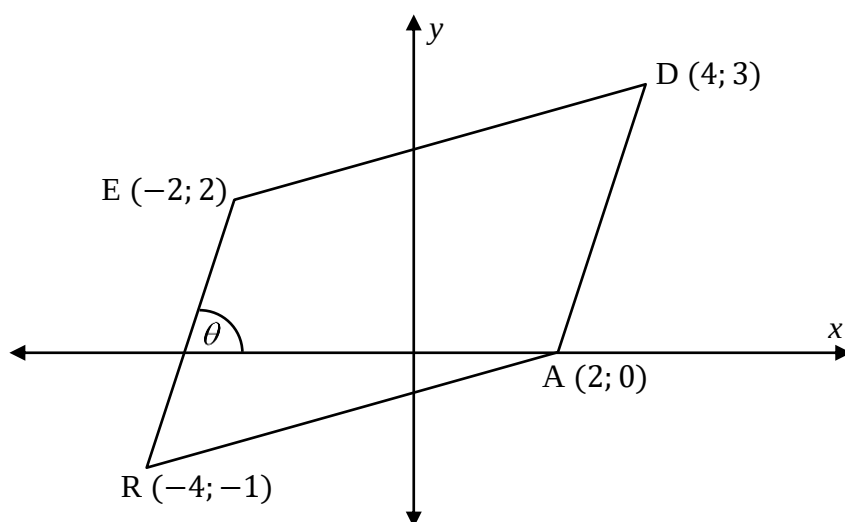
[7]

QUESTION 4

The inter passes amongst four football players in a premier league match were noted and then modelled on a Cartesian plane.



The inter passes form a quadrilateral with vertices $D(4; 3)$; $A(2; 0)$; $R(-4; -1)$ and $E(-2; 2)$ as represented on the diagram below. The angle of inclination of RE is θ .



- 4.1 Calculate the length of RE . Leave your answer in simplified surd form. (2)
- 4.2 Show that quadrilateral $DARE$ is a parallelogram. (3)
- 4.3 Determine the equation of DA in the form $y = mx + c$. (3)
- 4.4 Determine the magnitude of angle θ rounded off to the nearest degree. (2)

[10]

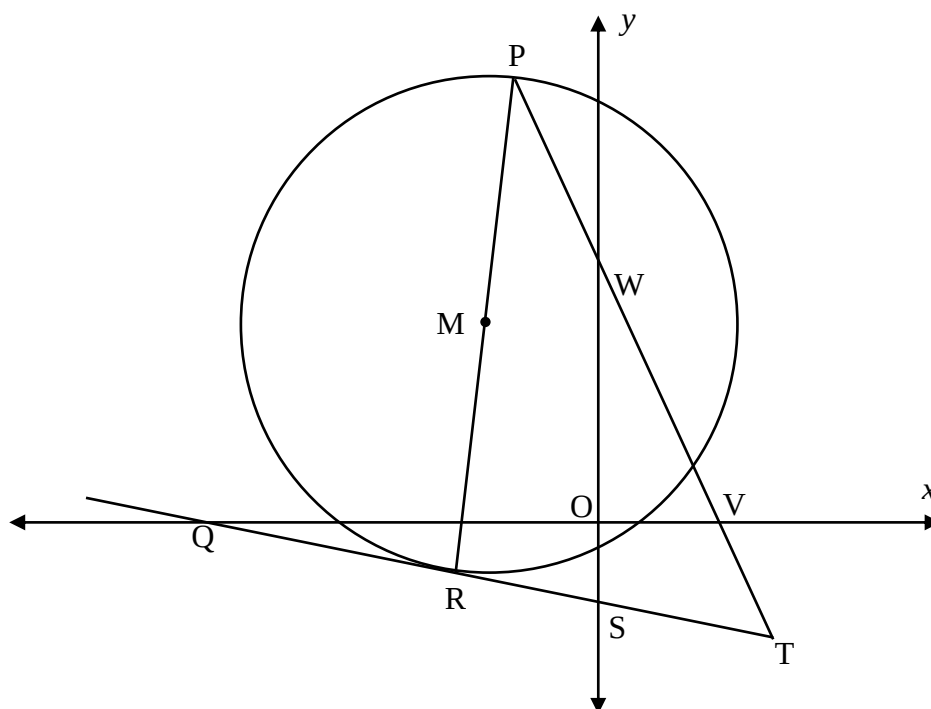
QUESTION 5

In the diagram, the equation of the circle with centre M is $(x+3)^2 + (y-4)^2 = 26$.

PR is a diameter of the circle. The equation of the tangent QT to the circle at R

is $y = -\frac{1}{5}x + k$. Q and S are the x- and y-intercepts of QT respectively.

PWVT is a straight line, with y-intercept W and x-intercept V.

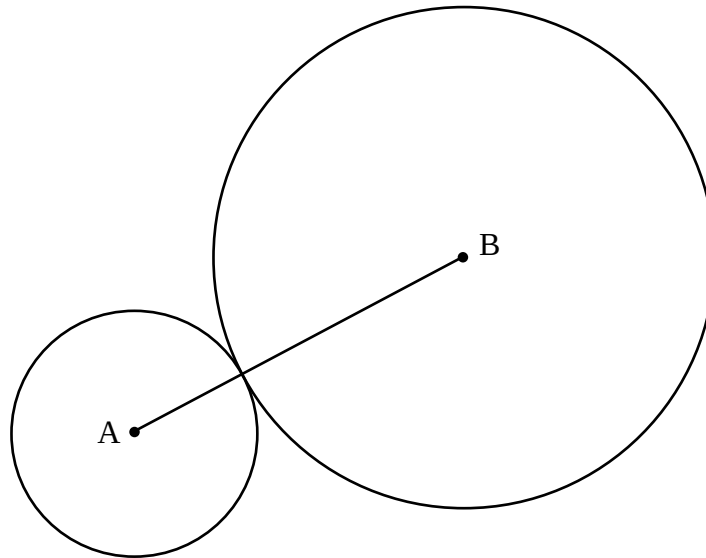


- 5.1 Write down the coordinates of M. (2)
- 5.2 Determine the length of the PM. (1)
- 5.3 Determine the equation of the diameter PMR. (3)
- 5.4 Determine the coordinates of R. (6)
- 5.5 Calculate the value of k . (2)
- 5.6 If $\widehat{OV} = 11,31^\circ$, prove that WVSQ is a cyclic quadrilateral. (4)

[18]

QUESTION 6

The diagram below represents two circles that intersect externally at only one point. The equation of circle with centre A is $x^2 + y^2 - 2x - 4y = 4$ and the equation of the circle with centre B is $(x - a)^2 + (y - 14)^2 = 100$, where $a > 0$.



- 6.1 Find the centre and radius of circle A. (4)
- 6.2 Determine the value of a . (5)
- [9]**

QUESTION 7

7.1 Simplify each of the following without the use of a calculator:

7.1.1 $\cos(70^\circ + \theta)\cos(10^\circ + \theta) + \sin(70^\circ + \theta)\sin(10^\circ + \theta)$ (3)

7.1.2
$$\frac{-2\sin\theta\cos\theta}{\cos^2(180^\circ - \theta) + \cos(90^\circ - \theta)\sin(\theta - 180^\circ)}$$
 (6)

7.2 If $\cos 61^\circ = p$, without using a calculator, find the values of the following in terms of p :

7.2.1 $\tan(-421^\circ)$ (3)

7.2.2 $\sin 122^\circ$ (3)

7.3 Given: $\cos(M - N) = \cos M \cos N + \sin M \sin N$

Use the formula for $\cos(M - N)$ to derive a formula for $\cos(M + N)$. (3)

7.4 Consider the identity:

$$\frac{3\cos 2x + 3\cos^2 x + 9\sin^2 x}{4 - 4\sin^2 x} = \frac{3}{2\cos^2 x}$$

7.4.1 Prove the identity. (5)

7.4.2 For which value of $x \in [0^\circ; 180^\circ]$ is this identity undefined? (2)

7.5 Find the general solution of the equation below:

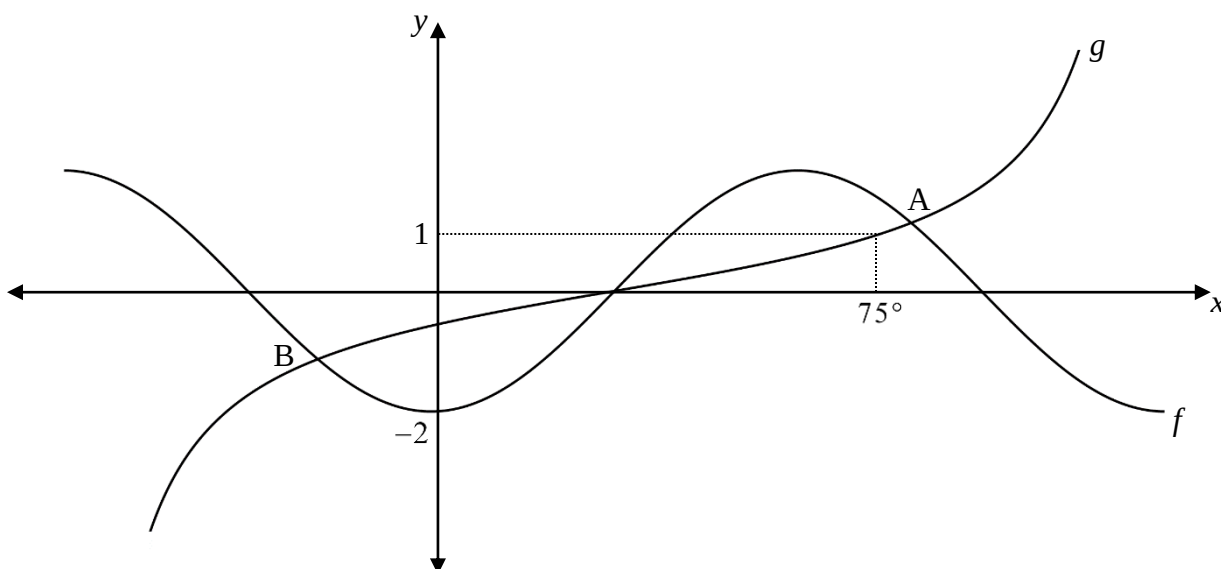
$$\sin x \cos 20^\circ + \cos x \sin 20^\circ = \sin 3x$$
 (5)

[30]

QUESTION 8

In the diagram below, the graphs f and g where $f(x) = a \cos bx$ and $g(x) = \tan(x - c)$ have been sketched for $x \in [-60^\circ; 120^\circ]$. The graph of g passes through the point $(75^\circ; 1)$.

A and B are points of intersection of f and g .



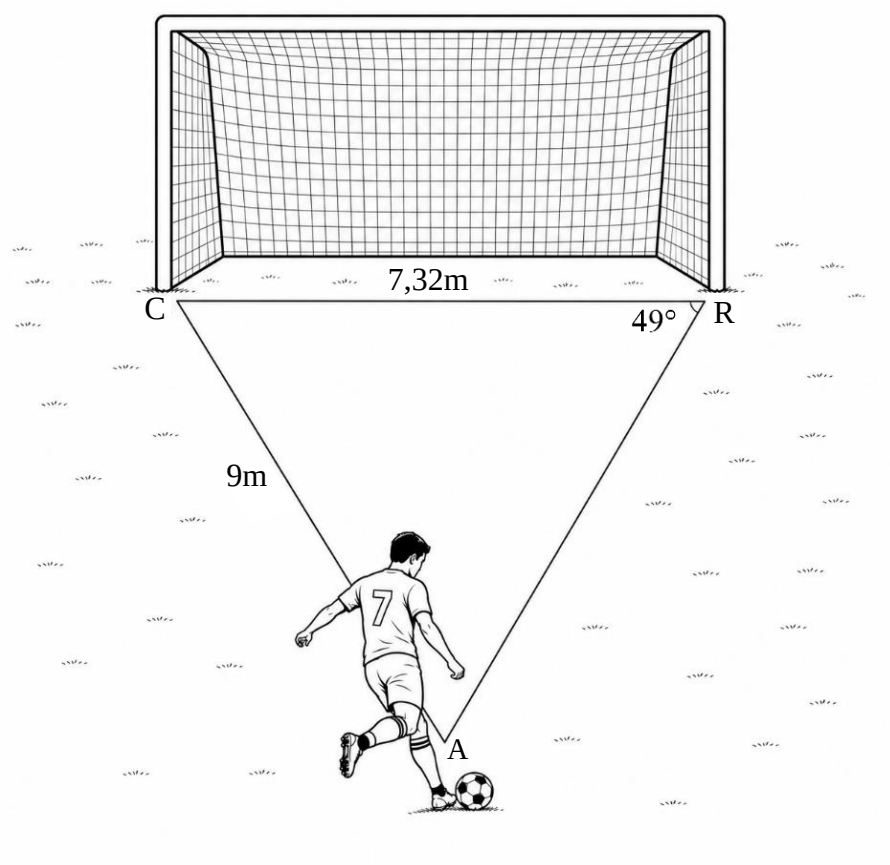
- 8.1 Determine the values of a , b and c . (3)
- 8.2 What is the period of f ? (1)
- 8.3 Write down the equation of an asymptote of g for the given interval. (1)
- 8.4 Given that the one general solution of $f(x) = g(x)$ is $x = 78,5^\circ + k \cdot 97^\circ; k \in \mathbb{Z}$, determine the coordinates of B. (2)
- 8.5 Determine the value(s) of x for which:
 - 8.5.1 $f(x) = 2$ (1)
 - 8.5.2 $f(x) \cdot g(x) \geq 0$ (1)

[9]

QUESTION 9

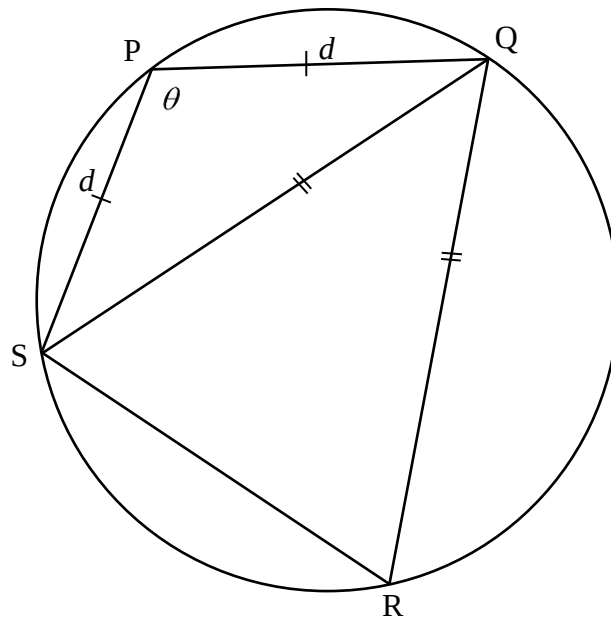
- 9.1 A penalty kick is one of the most challenging skills in soccer, requiring precise shooting technique and strong composure under pressure. It demands extensive planning and practice to strike the ball at an angle that give the goalkeeper little chance to save it.

A football player takes a shot on a standard net that is 7,32 m wide as illustrated in the diagram below. The player is 9 m from goalpost C and at an angle of 49° from goalpost R.



Calculate the angle A, to the nearest degree, that a goal can be shot. (3)

- 9.2 In the diagram below, PQRS is a cyclic quadrilateral with $QS = QR$ and $PQ = PS = d$ units. $\angle PSQ = \theta$.



Prove that:

9.2.1 $QS = d\sqrt{2 - 2\cos\theta}$ (3)

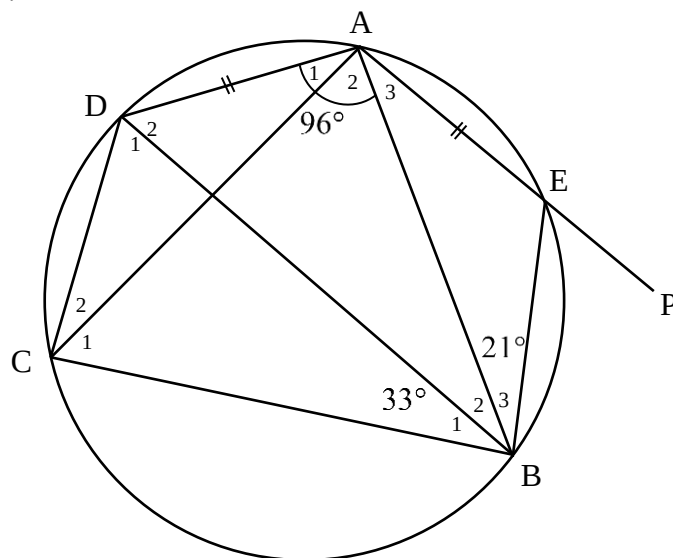
9.2.2 the area of $\triangle QRS = -d^2 \sin 2\theta(1 - \cos\theta)$ (4)

[10]

Provide reasons for your statements in QUESTIONS 10, 11, 12 and 13.

QUESTION 10

In the diagram below, ABCD is a cyclic quadrilateral with $\hat{BAD} = 96^\circ$ and $\hat{B}_1 = 33^\circ$. The diagonals AC and BD are drawn. E is a point on the circle and AE is produced to P. $\hat{ABE} = 21^\circ$ and $AD = AE$.



10.1 Write down, stating reasons, TWO other angles equal to 21° . (2)

10.2 Determine, with reasons, the size of $\hat{C}_1$. (3)

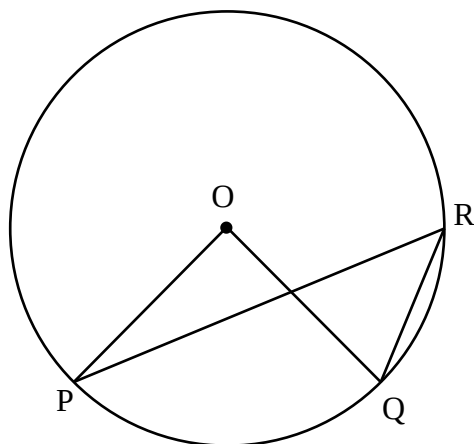
10.3 Determine, with reasons, the size of $\hat{BEP}$. (2)

10.4 Prove that $CB = AB$. (3)

[10]

QUESTION 11

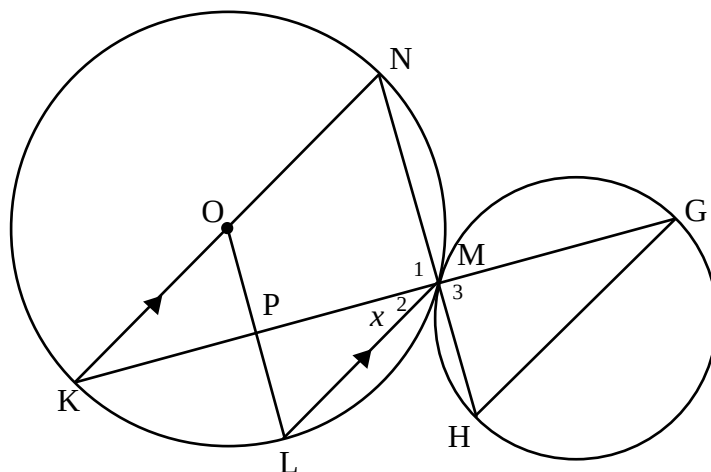
- 11.1 In the following sketch, O is the centre of the circle. P, Q and R are three points on the circumference of the circle.



Use the diagram above to prove the theorem which states that the angle at the centre of a circle is twice an angle at the circumference subtended by the same arc, that is prove that $\hat{POQ} = 2\hat{PRQ}$.

(5)

- 11.2 In the sketch below, O is the centre of the larger circle. Points K, N, M and L lie on the circumference of the larger circle. A smaller circle, passing through points M, G and H, touches the larger circle externally at M. KM intersects OL at P. Furthermore, $OK \parallel ML$ and $\hat{M}_2 = x$.



11.2.1 Express $\hat{L}$ in terms of x .

(4)

11.2.2 Prove that GH is the diameter of the smaller circle.

(3)

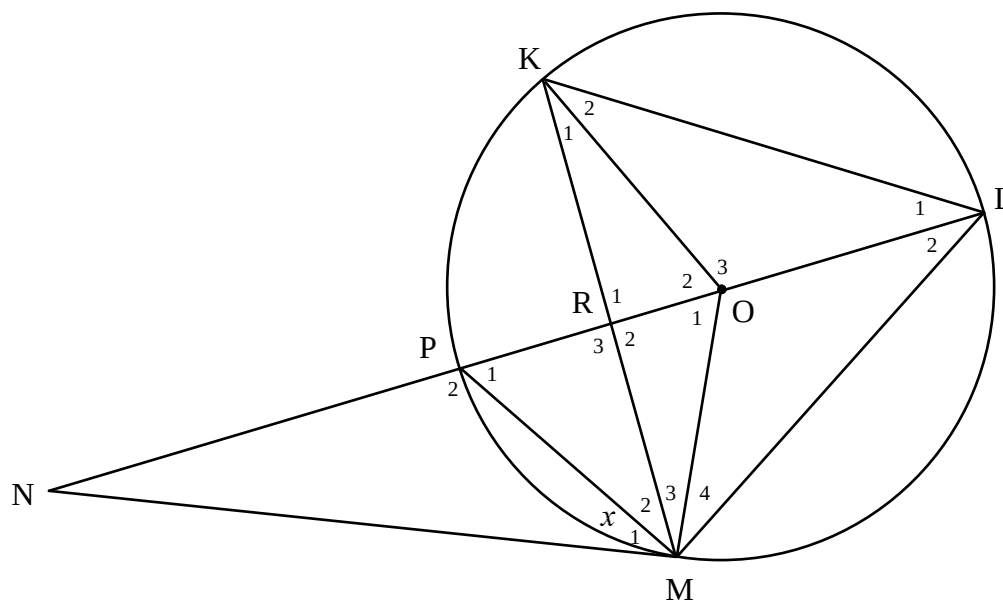
11.2.3 Determine the value of x for which OL bisects KM.

(3)

[15]

QUESTION 12

In the sketch below, O is the centre of the circle. K, L, M and P are points on the circumference of the circle. MN is a tangent to the circle at M. The diameter PL is extended to meet MN at N. KM and LP intersect at R. $\hat{M}_1 = x$.



12.1 Write down two angles equal to 90° . (2)

12.2 Write down the following angles in terms of x , giving reasons:

(a) $\hat{L}_2$ (2)

(b) $\hat{M}_4$ (2)

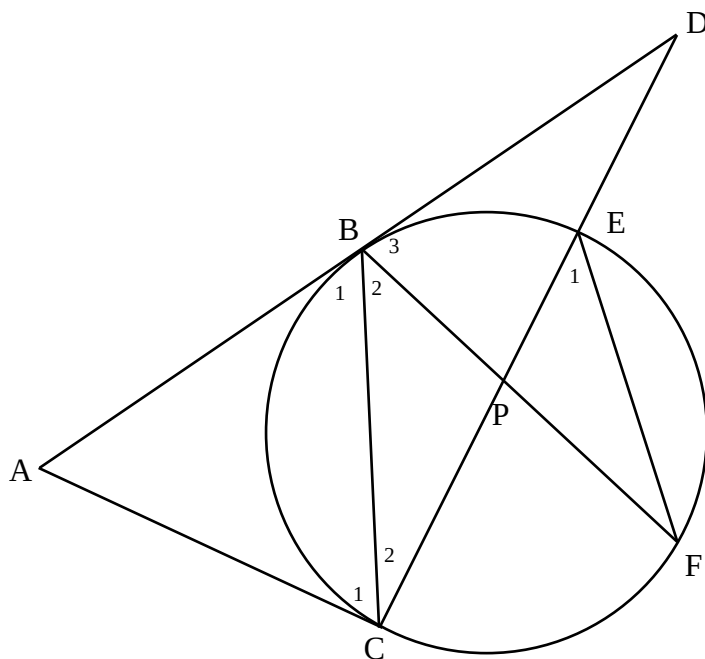
(c) $\hat{N}$ (3)

12.3 Join NK which is a tangent to the circle, and hence prove that KOMN is a cyclic quadrilateral. (2)

[11]

QUESTION 13

In the following sketch, ABD and AC are tangents to circle BEFC:



13.1 Prove that $\hat{C}_1 = \hat{D} + \hat{F}$. (4)

13.2 If $\hat{E}_1 = \hat{D}$, prove that BC is a tangent to circle BDP at B. (3)

[7]